

⚡ ENTERPRISE OPERATIONS GUIDE

Construction Inventory & Financial Suite

A practical, step-by-step master operational guide to multi-site material logistics, 3-way match fraud radar, and double-entry real estate accounting.

Application	Document Class	Effective Date	Target Audience
NAW PropertyFlow CRM	Standard Operating Procedure	August 2026	Board of Directors & Executives

1 The Business Value & Problem Solved

 **Executive Summary:** Real estate development firms across Nigeria lose between **15% to 25% of their construction budget** due to untracked site material diversion, ghost deliveries, double-invoicing, supplier price inflation, and delays in paper-based bookkeeping. The **NAW PropertyFlow Construction Inventory & Financial Suite** connects physical site operations directly to corporate double-entry accounting with 100% digital audit trails.

3-Way Match Fraud Radar

Cross-references Purchase Orders (PO), Goods Received Notes (GRN), and Supplier Invoices. No payment is released if price, quantity, or delivery proof differs by even ₦1.00.

GPS Geofenced Site Deliveries

Storekeepers cannot log deliveries remotely. The mobile GPS engine requires physical presence within a 300-meter radius of the construction yard.

Automated General Ledger

Every delivery, material issuance, or vendor payment creates balanced Debit/Credit journal postings directly in the corporate Chart of Accounts.

Dynamic Payment Plan Calculator

Automatic interest calculation across 3M, 6M, 9M, 12M+ tenures with itemized VAT (7.5%), tax, and digitally stamped PDF receipts.

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Stakeholder Persona Matrix

The platform enforces strict Role-Based Access Control (RBAC) across 6 key roles:

ROLE / PERSONA	PRIMARY RESPONSIBILITY	ANTI-FRAUD & AUDIT CONTROLS
1. Site Engineer	Submits Material Requisitions (MRF) for specific building milestones.	Cannot exceed QS Bill of Materials (BOM) without variance approval.
2. Quantity Surveyor (QS)	Sets material ratios, checks market price benchmarks, and creates POs.	Audited against regional price benchmarks (Abuja Dei-Dei / Lagos Orile).
3. Managing Director (MD)	Authorizes high-value Purchase Orders and corporate disbursements.	Tiered authorization: 1-click approval for POs exceeding ₦5M / ₦20M.
4. Site Storekeeper	Inspects delivered goods, logs batches, and issues materials on site.	GPS Geofence locked (300m radius) + 90-day cement shelf-life tracker.
5. Financial Controller	Audits 3-way matching, approves payouts, and monitors balance sheets.	Zero manual data entry; automated balanced journal generation.
6. Verified Supplier	Uploads digital PDF invoices and tracks payment status.	Access limited strictly to private Supplier Portal with PO matching.

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End-to-End Site Procurement Flow (Step-by-Step)

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Step 1: Bill of Materials (BOM) Standard Definition

QUANTITY
SURVEYOR

The Quantity Surveyor establishes the engineering standard for each structural activity before building begins.

Example (Grade 25 Concrete Slab – 1 m³):

- 7.0 Bags Dangote Falcon Cement (50kg) @ ₦8,500/bag
- 0.5 Trips Coarse River Sand @ ₦120,000/trip
- 0.8 Trips 3/4 Clean Granite Stone @ ₦280,000/trip
- Allowable Site Tolerance: **±5.00%**

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Step 2: Material Requisition Form (MRF) Submission

SITE ENGINEER

When work begins on the *1st Floor Suspended Slab Casting* at Hutu Prestige Estate, the Site Engineer logs into the mobile portal and requests **500 bags of cement**.

- The system verifies that 500 bags match the remaining casting volume ($71.4 \text{ m}^3 \times 7 \text{ bags} = 500$).
- If the engineer enters `700 bags`, the system blocks submission with an **Over-Allocation Red Alert (+40%)** requiring civil justification.

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Step 3: Purchase Order (PO) & Tiered Authorization

QS & MANAGING
DIRECTOR

The QS converts the approved MRF into an official **Purchase Order (PO-2026-0042)** assigned to *Dangote Building Solutions Ltd.*

- **Price Benchmark Check:** Quoted ₦8,500/bag is compared with Dei-Dei Market index (₦8,500). Variance = 0%.
- **Total Value:** $500 \times ₦8,500 = ₦4,250,000.00$.
- **Approval Tier:** Under ₦5,000,000 threshold $\rightarrow$ Authorized instantly by QS/Project Manager.

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Step 4: GPS Geofenced Goods Receipt (GRN) on Site

**SITE
STOREKEEPER**

The supplier's flatbed truck arrives at Katampe Extension Site Yard.

- **Geofence Verification:** The Storekeeper's mobile browser confirms GPS presence within 300m of Plot 402 Katampe Extension (9.0820°N , 7.4810°E).
- **Batch & Quality Inspection:** Physical tally (500 bags), Batch No: DAN-ABJ-882, Live photo of stacked pallets uploaded.
- **Shelf-Life Countdown:** System initiates a 90-day timer for cement freshness.

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Step 5: Material Issuance (MIV) & Wastage Tracking

**STOREKEEPER & SUB-
CONTRACTOR**

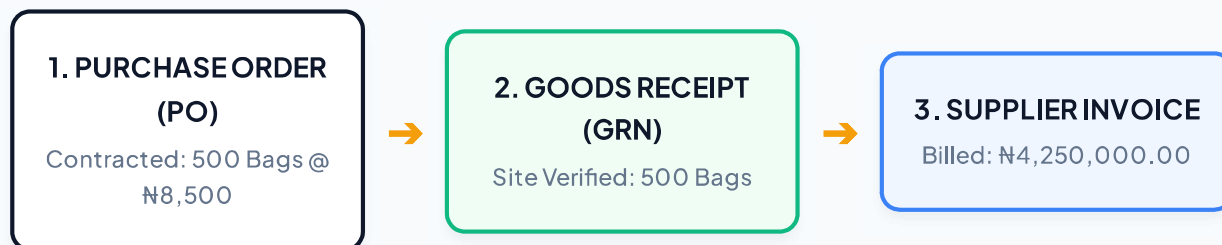
When site masons withdraw 200 bags for today's casting, the Storekeeper generates a digital **Material Issue Voucher (MIV)**.

- Site stock automatically reduces from 500 bags to 300 bags in real time.
- If damaged bags exceed standard site tolerance (1.5x multiplier), an automated ****Wastage Alert**** is routed to Project Directors.

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Financial Settlement & 3-Way Match Fraud Radar

THE 3-WAY MATCHING RECONCILIATION RADAR



✓ Quantities Match (100%) • ✓ Unit Price Matches (100%) • ✓ GPS Geofence Verified → PAYMENT CLEARED



Double-Entry General Ledger Automation

FINANCIAL CONTROLLER

Upon releasing supplier payment via bank transfer, the system automatically posts balanced journals without manual bookkeeping:

// 1. Recording Materials Received into Site Assets

[DEBIT] Account #1300 – Site Materials Inventory (Current Asset)

₦4,250,000.00

[CREDIT] Account #1010 – Corporate Operations Bank Account (Asset)

₦4,250,000.00

// 2. When Materials Are Consumed On 1st Floor Slab (MIV Issuance)

[DEBIT] Account #5100 – Direct Project Construction Cost (COGS)

₦1,700,000.00

[CREDIT] Account #1300 – Site Materials Inventory (Current Asset)

₦1,700,000.00

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Client Sales, Dynamic Spread & Official PDF Receipts

Configurable installment spreads with live interest calculations and transparent statutory breakdowns:

Tenure Option	Interest Spread Surcharge	Initial Commitment	Statutory VAT & Tax Breakdown
Outright (0 Months)	0.00% (No Interest)	100% Full Payment	Itemized on PDF Receipt
3 Months Spread	0.00% (Interest Free)	30% Deposit	Itemized on PDF Receipt
6 Months Spread	+5.00% Surcharge	25% Deposit	Itemized on PDF Receipt
9 Months Spread	+7.50% Surcharge	20% Deposit	Itemized on PDF Receipt
12 Months Spread	+10.00% Surcharge	20% Deposit	Itemized on PDF Receipt
18 - 24 Months	+15.00% - +20.00%	15% Deposit	Itemized on PDF Receipt



Official Digitally Stamped PDF Receipt Structure:

When an investor deposits funds for any unit (e.g. ₦50M Duplex on a 6-month plan), the PDF receipt displays: **Base Price (₦50,000,000) + Tenure Interest (+₦2,500,000) + VAT 7.5% (+ ₦3,750,000) = Total Contract Sum (₦56,250,000)**, along with Cumulative Paid and Remaining Balance.